

CA1: Dataframe Manipulation with Spotify Data

Introduction

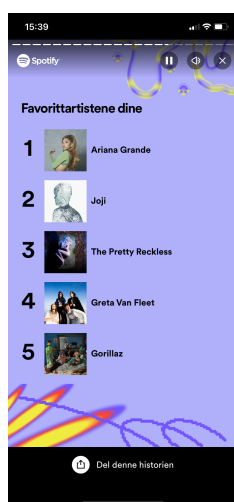
Pandas is an extremely powerful tool to handle large amounts of tabular data. In this compulsory assignment, you will use Pandas to explore one of the TA's personal spotify data in depth.

Additional information:

- Feel free to create additional code cells if you feel that one cell per subtask is not sufficient.
- Remember, Pandas uses very efficient code to handle large amounts of data. For-loops are not efficient. If you ever have to use a for-loop to loop over the rows in the DataFrame, you have *probably* done something wrong.
- Label all graphs and charts if applicable.

Task

I typically enjoy indie and rock music. I am a big fan of everything from old-fashioned rock and roll like Led Zeppelin and Jimi Hendrix, to newer indie artists like Joji and Lana Del Rey. This is why my spotify wrapped for 2023 came as quite a surprise:



Now, I'm no hater of pop music, but this was unexpected. For this assignment, you will investigate my listening habits, including a deep dive into my Ariana Grande listening habits, and try to find an answer to why she was my top artist; was there a fault in the spotify

algorithm? Am I actually secretly an *Arianator*? (yes, I did have to look that up). Or am I just lying to myself about how often I listen to guilty pleasure music?

Part 1: Initial loading and exploration

1.0 Import necessary libraries:

pandas, numpy, matplotlib.pyplot (other libraries such as seaborn or plotly are also allowed if you want prettier plots). It might also be a good idea to use **os** for task 2.0

```
In [2]: import matplotlib.pyplot as plt
# ---- Insert other imports ----
import pandas as pd
```

1.1 Loading the data

Load the dataset in the file `streaming_history_0.csv` into a Pandas DataFrame called `df_spotify_0`.

```
In [3]: df_spotify_0 = pd.read_csv("spotify_data\streaminghistory0.csv")
```

1.2 Help function

Use the Python command `help` to help you understand how to use the `pd.DataFrame.head` and `pd.DataFrame.tail` methods.

```
In [4]: help(pd.DataFrame.head)
```

Help on function head in module pandas.core.generic:

```
head(self, n: 'int' = 5) -> 'Self'
    Return the first `n` rows.
```

This function returns the first `n` rows for the object based on position. It is useful for quickly testing if your object has the right type of data in it.

For negative values of `n`, this function returns all rows except the last `|n|` rows, equivalent to ``df[:n]``.

If n is larger than the number of rows, this function returns all rows.

Parameters

n : int, default 5
Number of rows to select.

Returns

same type as caller
The first `n` rows of the caller object.

See Also

DataFrame.tail: Returns the last `n` rows.

Examples

```
>>> df = pd.DataFrame({'animal': ['alligator', 'bee', 'falcon', 'lion',
...                               'monkey', 'parrot', 'shark', 'whale', 'zebra']})
>>> df
   animal
0  alligator
1      bee
2   falcon
3    lion
4   monkey
5   parrot
6    shark
7   whale
8    zebra
```

Viewing the first 5 lines

```
>>> df.head()
   animal
0  alligator
1      bee
2   falcon
3    lion
4   monkey
```

Viewing the first `n` lines (three in this case)

```
>>> df.head(3)
      animal
0  alligator
1        bee
2       falcon
```

For negative values of `n`

```
>>> df.head(-3)
      animal
0  alligator
1        bee
2       falcon
3        lion
4       monkey
5       parrot
```

```
In [5]: help(pd.DataFrame.tail)
```

Help on function tail in module pandas.core.generic:

```
tail(self, n: 'int' = 5) -> 'Self'
    Return the last `n` rows.
```

This function returns last `n` rows from the object based on position. It is useful for quickly verifying data, for example, after sorting or appending rows.

For negative values of `n`, this function returns all rows except the first `|n|` rows, equivalent to ``df[|n|:]``.

If n is larger than the number of rows, this function returns all rows.

Parameters

n : int, default 5
Number of rows to select.

Returns

type of caller
The last `n` rows of the caller object.

See Also

DataFrame.head : The first `n` rows of the caller object.

Examples

```
>>> df = pd.DataFrame({'animal': ['alligator', 'bee', 'falcon', 'lion',
...                               'monkey', 'parrot', 'shark', 'whale', 'zebra']})
>>> df
   animal
0  alligator
1      bee
2   falcon
3    lion
4   monkey
5   parrot
6    shark
7   whale
8    zebra
```

Viewing the last 5 lines

```
>>> df.tail()
   animal
4  monkey
5  parrot
6   shark
7   whale
8   zebra
```

Viewing the last `n` lines (three in this case)

```
>>> df.tail(3)
      animal
6  shark
7  whale
8  zebra
```

For negative values of `n`

```
>>> df.tail(-3)
      animal
3  lion
4  monkey
5  parrot
6  shark
7  whale
8  zebra
```

1.3 Getting an overview

Print the first **five** and last **ten** rows of the dataframe. Have a quick look at which columns are in the dataset.

```
In [6]: df_spotify_0.head(5)
```

```
Out[6]:
```

	endTime	artistName	trackName	msPlayed
0	2022-12-03 02:02	Cigarettes After Sex	Truly	30000.0
1	2022-12-03 02:02	Leonard Cohen	Take This Waltz - Paris Version	8210.0
2	2022-12-06 21:05	Vlad Holiday	So Damn Into You	37895.0
3	2022-12-06 21:05	Lorde	Team	8984.0
4	2022-12-06 21:05	Ariana Grande	Into You	1221.0

```
In [7]: df_spotify_0.tail(10)
```

Out[7]:

	endTime	artistName	trackName	msPlayed
11949	2023-01-02 20:58	Ariana Grande	six thirty	1699.0
11950	2023-01-02 20:58	Leonard Cohen	Thanks for the Dance	19483.0
11951	2023-01-02 20:59	Des Rocs	Used to the Darkness	185.0
11952	2023-01-02 20:59	Caroline Polachek	Hit Me Where It Hurts	603.0
11953	2023-01-02 20:59	Caroline Polachek	Hit Me Where It Hurts	208.0
11954	2023-01-02 20:59	Kaizers Orchestra	Resistansen	208.0
11955	2023-01-02 20:59	Mr.Kitty	After Dark	101447.0
11956	2023-01-02 20:59	daddy's girl	after dark x sweater weather	301.0
11957	2023-01-02 20:59	daddy's girl	after dark x sweater weather	208.0
11958	2023-01-02 20:59	daddy's girl	after dark x sweater weather	789.0

1.4 Formatting correctly

When working with Pandas, it's very useful to have columns which contains dates in a specific format called *datetime*. This allows for efficient manipulation and analysis of time-series data, such as sorting, filtering by date or time, and resampling for different time periods. Figure out which column(s) would be appropriate to convert to datetime, if any, and if so, perform the conversion to the correct format.

```
In [8]: # Endtime is the only appropriate column in the dataset
df_spotify_0["dateTime"] = pd.to_datetime(df_spotify_0["endTime"])
```

1.5 Unique artists

Find how many unique artists are in the dataset.

```
In [9]: df_spotify_0["artistName"].nunique()
```

Out[9]: 495

1.6 Unique songs

Find how many unique songs are in the dataset.

```
In [10]: df_spotify_0["trackName"].nunique()
```

Out[10]: 1308

Part 1: Questions

Q1: Which columns are in the dataset?

Q2: What timeframe does the dataset span?

Q3: How many unique artists are in the dataset?

Q4: How many unique songs are in the dataset?

Part 2: Working with all the data

2.0 Importing all the dataframes

In Task 1, you only worked with about a month worth of data. Now, you will work with over a year worth.

In the *spotify_data* folder, there is more than just one listening record. Load each of the 14 listening records into a dataframe (1 dataframe per listening record), and concatenate them together into one large dataframe named `df`.

```
In [11]: df_spotify_0 = pd.read_csv("spotify_data\streaminghistory0.csv")
df_spotify_1 = pd.read_csv("spotify_data\streaminghistory1.csv")
df_spotify_2 = pd.read_csv("spotify_data\streaminghistory2.csv")
df_spotify_3 = pd.read_csv("spotify_data\streaminghistory3.csv")
df_spotify_4 = pd.read_csv("spotify_data\streaminghistory4.csv")
df_spotify_5 = pd.read_csv("spotify_data\streaminghistory5.csv")
df_spotify_6 = pd.read_csv("spotify_data\streaminghistory6.csv")
df_spotify_7 = pd.read_csv("spotify_data\streaminghistory7.csv")
df_spotify_8 = pd.read_csv("spotify_data\streaminghistory8.csv")
df_spotify_9 = pd.read_csv("spotify_data\streaminghistory9.csv")
df_spotify_10 = pd.read_csv("spotify_data\streaminghistory10.csv")
df_spotify_11 = pd.read_csv("spotify_data\streaminghistory11.csv")
df_spotify_12 = pd.read_csv("spotify_data\streaminghistory12.csv")
df_spotify_13 = pd.read_csv("spotify_data\streaminghistory13.csv")

df = pd.concat([
    df_spotify_0,
    df_spotify_1,
    df_spotify_2,
    df_spotify_3,
    df_spotify_4,
    df_spotify_5,
    df_spotify_6,
    df_spotify_7,
    df_spotify_8,
    df_spotify_9,
    df_spotify_10,
    df_spotify_11,
    df_spotify_12,
    df_spotify_13,
])
```

2.1 Sorting by time

Datasets often aren't perfect. One example of an issue that could occur is that the time-based data might not be in chronological order. If this were to happen, the rows in your dataframe could be in the wrong order. To ensure this isn't an issue in your dataframe, you should sort the dataframe in chronological order, from oldest to newest.

```
In [12]: df["dateTime"] = pd.to_datetime(df["endTime"])
df = df.sort_values(by="dateTime")
```

2.2 Setting a timeframe

For this investigation, we are only interested in investigating listening patterns from **2023**. Remove any data not from **2023** from the DataFrame.

```
In [13]: df = df[df["dateTime"].dt.year == 2023]
print(df["dateTime"])

# print(df[df["artistName"] == "Rage Against The Machine"])
```

```
10881    2023-01-01 01:17:00
10882    2023-01-01 01:17:00
10883    2023-01-01 01:17:00
10884    2023-01-01 01:17:00
10885    2023-01-01 01:17:00
...
11962    2023-12-07 21:13:00
11968    2023-12-07 21:13:00
11970    2023-12-07 21:14:00
11969    2023-12-07 21:14:00
11971    2023-12-07 21:17:00
Name: dateTime, Length: 156557, dtype: datetime64[ns]
```

2.3 Deleting rows

Often in Data Science, you will encounter when a row entry has the value *NaN*, indicating missing data. These entries can skew your analysis, leading to inaccurate conclusions. For this task, identify and remove any rows in your DataFrame that contain NaN values.

Later in the course, you might encounter other techniques of dealing with missing data, typically referred to as *data imputation*. Here, though, you are just supposed to delete the entire rows with missing data.

```
In [14]: df = df.dropna()
```

2.4 Convert from milliseconds to seconds

From `msPlayed`, create a new column `secPlayed` with the data converted from milliseconds to seconds. Then delete the column `msPlayed`.

```
In [15]: df["secPlayed"] = df["msPlayed"] / 1000
df = df.drop(columns=["msPlayed"])
```

2.5 Finding top 10 favorite artists

Find the top `ten` artists with the highest total play time (in seconds). Plot your findings in a bar graph.

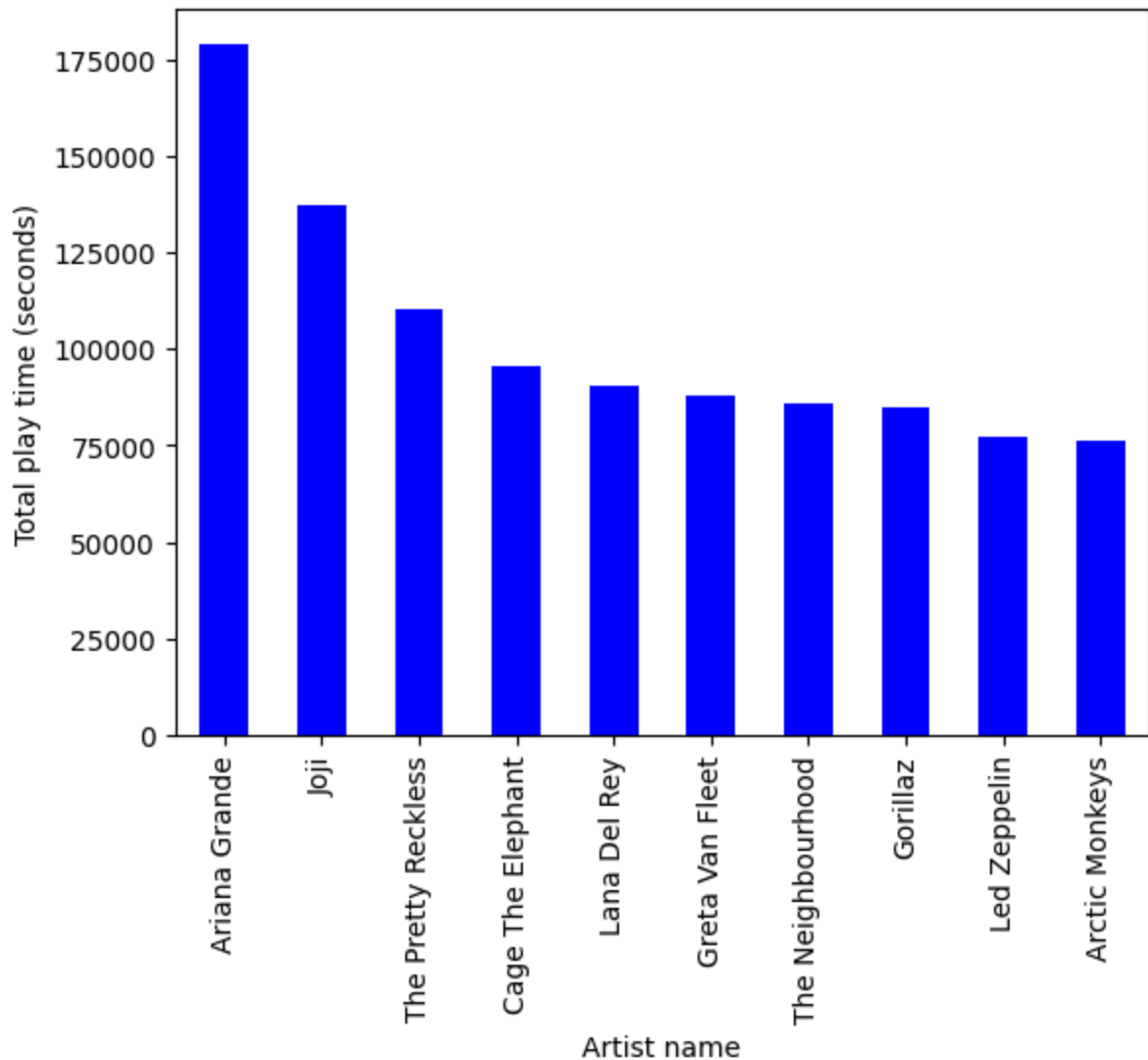
(hint: start by creating a new DataFrame with only `artistName` and your time column. To proceed, you will also likely need the `groupby` command from Pandas.)

```
In [16]: artists_df = df.groupby('artistName')['secPlayed'].sum()

top_10_artists = artists_df.nlargest(10)

top_10_artists.plot(kind="bar", color="b")
plt.xlabel("Artist name")
plt.ylabel("Total play time (seconds)")
```

```
Out[16]: Text(0, 0.5, 'Total play time (seconds)')
```



2.6 Finding top 10 favorite songs

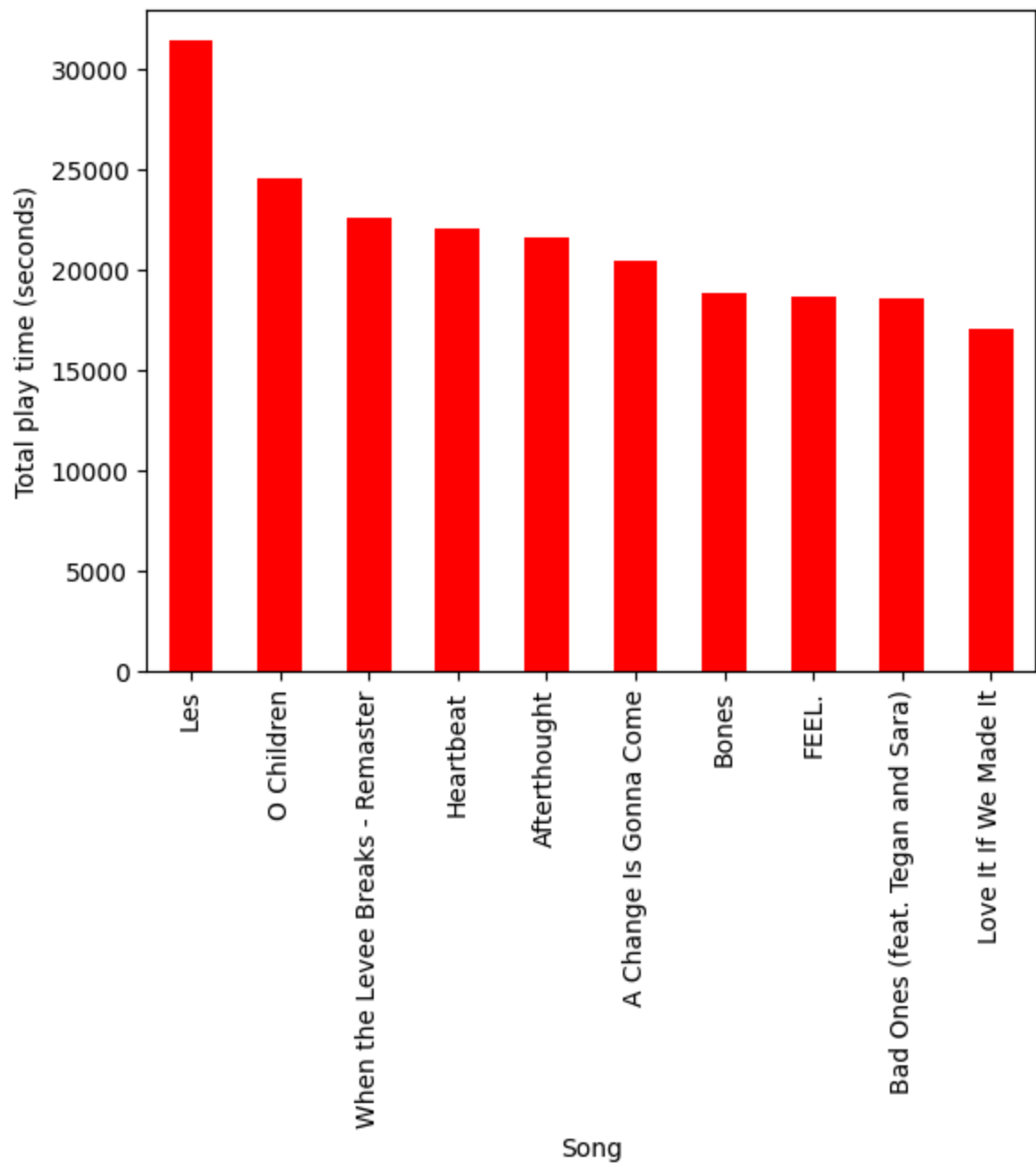
Find the top `ten` songs with the highest play time. Create a graph visualizing the results.

```
In [17]: songs_df = df.groupby("trackName")["secPlayed"].sum()

top_10_songs = songs_df.nlargest(10)

top_10_songs.plot(kind="bar", color="r")
plt.xlabel("Song")
plt.ylabel("Total play time (seconds)")
```

```
Out[17]: Text(0, 0.5, 'Total play time (seconds)')
```



Part 3: Further analysis

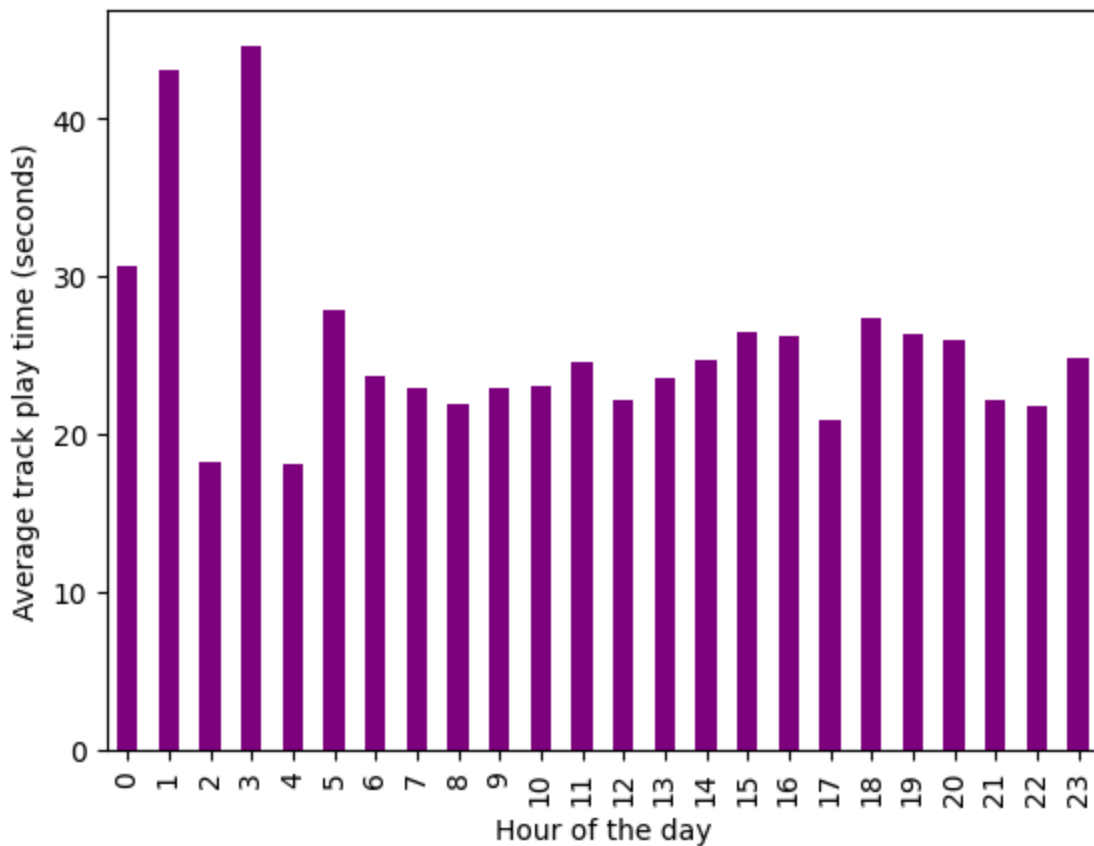
3.0 Average listening time by hour

Generate a plot that displays the average amount of time that music is played for each hour of the day.

```
In [18]: listening_Time_By_Hour = df.groupby(df["dateTime"].dt.hour)["secPlayed"].mean()

listening_Time_By_Hour.plot(kind="bar",color="purple")
plt.xlabel("Hour of the day")
plt.ylabel("Average track play time (seconds)")
```

```
Out[18]: Text(0, 0.5, 'Average track play time (seconds)')
```



3.1 Morning music and evening music

I think many people find that some types of music are more suitable for morning listening and some music is more suitable for evening listening. Create a plot that compares the play time of the artists *Leonard Cohen* and *Rage Against the Machine* on an hour-by-hour basis. See if there are any differences.

```
In [19]: artists = df[df["artistName"].isin(["Leonard Cohen", "Rage Against The Machine"])]

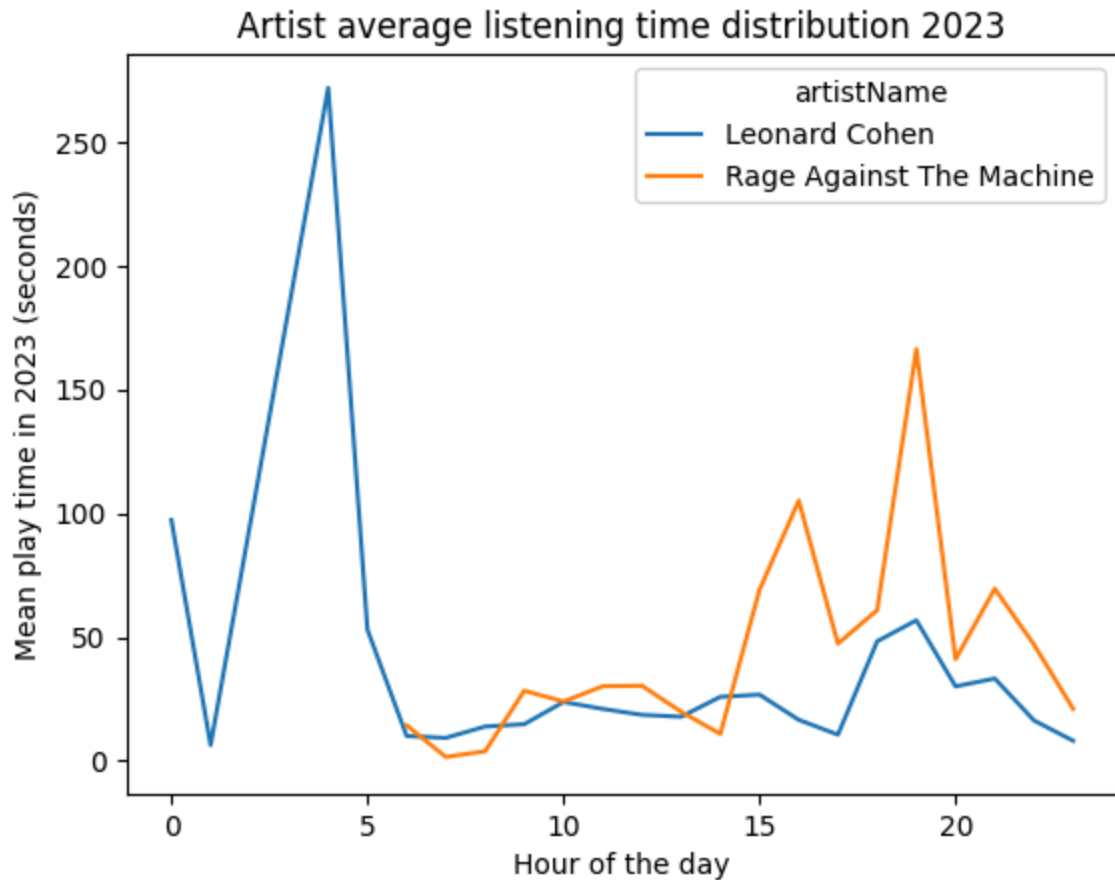
# Mean
```

```

artist_comparison_mean = artists.groupby([artists["dateTime"].dt.hour, artists["art
artist_comparison_mean.plot(kind="line")
plt.title("Artist average listening time distribution 2023")
plt.xlabel("Hour of the day")
plt.ylabel("Mean play time in 2023 (seconds)")

```

Out[19]: Text(0, 0.5, 'Mean play time in 2023 (seconds)')



3.2 Analysing skipped songs

Determining whether a song was skipped or listened to can be challenging. For this analysis, we'll simplify by defining a skipped song as any track played for less than 30 seconds.

Conversely, a song played for 30 seconds or more is considered listened to.

Add a column to your DataFrame to reflect this criteria: set the value to 1 if the song was played for less than 30 seconds (indicating a skipped song), and 0 if it was played for 30 seconds or longer.

```

In [20]: df["skipped"] = (df["secPlayed"] < 30).astype(int)
print(df)

```

	endTime	artistName	trackName \
10881	2023-01-01 01:17	Ariana Grande	7 rings
10882	2023-01-01 01:17	Ariana Grande	7 rings
10883	2023-01-01 01:17	Ariana Grande	positions
10884	2023-01-01 01:17	Peach Pit	Being so Normal
10885	2023-01-01 01:17	Kelly Clarkson	Santa, Can't You Hear Me
...	...	...	...
11962	2023-12-07 21:13	Arctic Monkeys	Snap Out Of It
11968	2023-12-07 21:13	Ariana Grande	off the table (with The Weeknd)
11970	2023-12-07 21:14	Leonard Cohen	Thanks for the Dance
11969	2023-12-07 21:14	Ariana Grande	my hair
11971	2023-12-07 21:17	The Vaccines	Your Love Is My Favourite Band

	dateTime	secPlayed	skipped
10881	2023-01-01 01:17:00	0.139	1
10882	2023-01-01 01:17:00	0.487	1
10883	2023-01-01 01:17:00	0.417	1
10884	2023-01-01 01:17:00	2.205	1
10885	2023-01-01 01:17:00	0.278	1
...	...	...	...
11962	2023-12-07 21:13:00	1.497	1
11968	2023-12-07 21:13:00	13.448	1
11970	2023-12-07 21:14:00	9.317	1
11969	2023-12-07 21:14:00	23.757	1
11971	2023-12-07 21:17:00	14.661	1

[156539 rows x 6 columns]

3.3 Plotting skipped songs

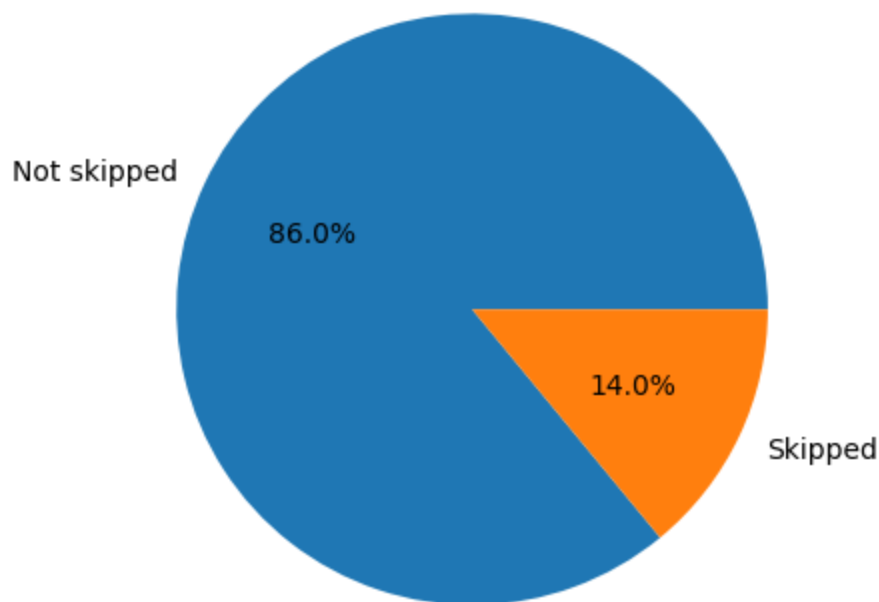
Create a pie-chart that compares amount of skipped songs to amount of non-skipped songs.

```
In [21]: skipped = df["skipped"].value_counts()

skipped.plot(kind="pie", labels=["Not skipped", "Skipped"], autopct="%1.1f%")
plt.title("Skipped VS Non skipped songs 2023")
plt.ylabel("")
```

```
Out[21]: Text(0, 0.5, '')
```

Skipped VS Non skipped songs 2023



3.4 Artists by percentage of songs skipped

For each artist in the dataset, calculate which percentage of their songs was skipped. Store this information in a new DataFrame called `df_skipped`. Store the percentage of skipped songs in a new column named `SkipRate`

Example: If an artist has **100** songs in your dataset and **25** of these were skipped, the percentage of skipped songs for this artist would be $\frac{25}{100} = 25\%$

```
In [22]: df_skipped = pd.DataFrame(df.groupby("artistName")["skipped"].mean())
df_skipped["skipRate"] = df_skipped["skipped"] * 100
print(df_skipped)
```

artistName	skipped	skipRate
10cc	0.678571	67.857143
2Pac	0.861598	86.159844
3 Doors Down	0.500000	50.000000
4 Non Blondes	0.721311	72.131148
50 Cent	0.678571	67.857143
...	...	...
squeeda	0.666667	66.666667
tenkousei.	1.000000	100.000000
trxxshed	0.500000	50.000000
xander.	0.375000	37.500000
Édith Piaf	0.941935	94.193548

[956 rows x 2 columns]

3.5 Comparing artists by skip-rate

Find the `three` top artists with the lowest skip-rate and the `three` with the highest. Print their names, along with their skip-rate.

```
In [23]: # lowest
least_skipped_artists = df_skipped["skipRate"].nsmallest(3)
print(least_skipped_artists, "\n")

# highest
most_skipped_artists = df_skipped["skipRate"].nlargest(3)
print(most_skipped_artists)
```

```
artistName
Gloria Gaynor      0.000000
Roc Boyz          11.111111
LACES             14.285714
Name: skipRate, dtype: float64
```

```
artistName
A Problem Squared  100.0
Acid Ghost         100.0
Albert Hammond Jr  100.0
Name: skipRate, dtype: float64
```

Part 4: God Is a Data Scientist - The Ariana Deep-Dive

4.0 Ariana-DataFrame:

Create a new DataFrame called `df_ariana`, containing only rows with music by Ariana Grande.

```
In [24]: df_ariana = df[df["artistName"] == "Ariana Grande"]

print(df_ariana)
```

		endTime	artistName	trackName \
10881	2023-01-01 01:17	Ariana Grande	7 rings	
10882	2023-01-01 01:17	Ariana Grande	7 rings	
10883	2023-01-01 01:17	Ariana Grande	positions	
10887	2023-01-01 01:17	Ariana Grande	Santa Baby	
10888	2023-01-01 01:17	Ariana Grande	Right There (feat. Big Sean)	
...	...	...	...	
11948	2023-12-07 17:46	Ariana Grande	Almost Is Never Enough	
11955	2023-12-07 20:51	Ariana Grande	needy	
11961	2023-12-07 21:13	Ariana Grande	pete davidson	
11968	2023-12-07 21:13	Ariana Grande	off the table (with The Weeknd)	
11969	2023-12-07 21:14	Ariana Grande	my hair	

	dateTime	secPlayed	skipped
10881	2023-01-01 01:17:00	0.139	1
10882	2023-01-01 01:17:00	0.487	1
10883	2023-01-01 01:17:00	0.417	1
10887	2023-01-01 01:17:00	12.293	1
10888	2023-01-01 01:17:00	22.929	1
...	...	...	...
11948	2023-12-07 17:46:00	28.483	1
11955	2023-12-07 20:51:00	26.220	1
11961	2023-12-07 21:13:00	0.603	1
11968	2023-12-07 21:13:00	13.448	1
11969	2023-12-07 21:14:00	23.757	1

[19337 rows x 6 columns]

4.1 Average skip rate

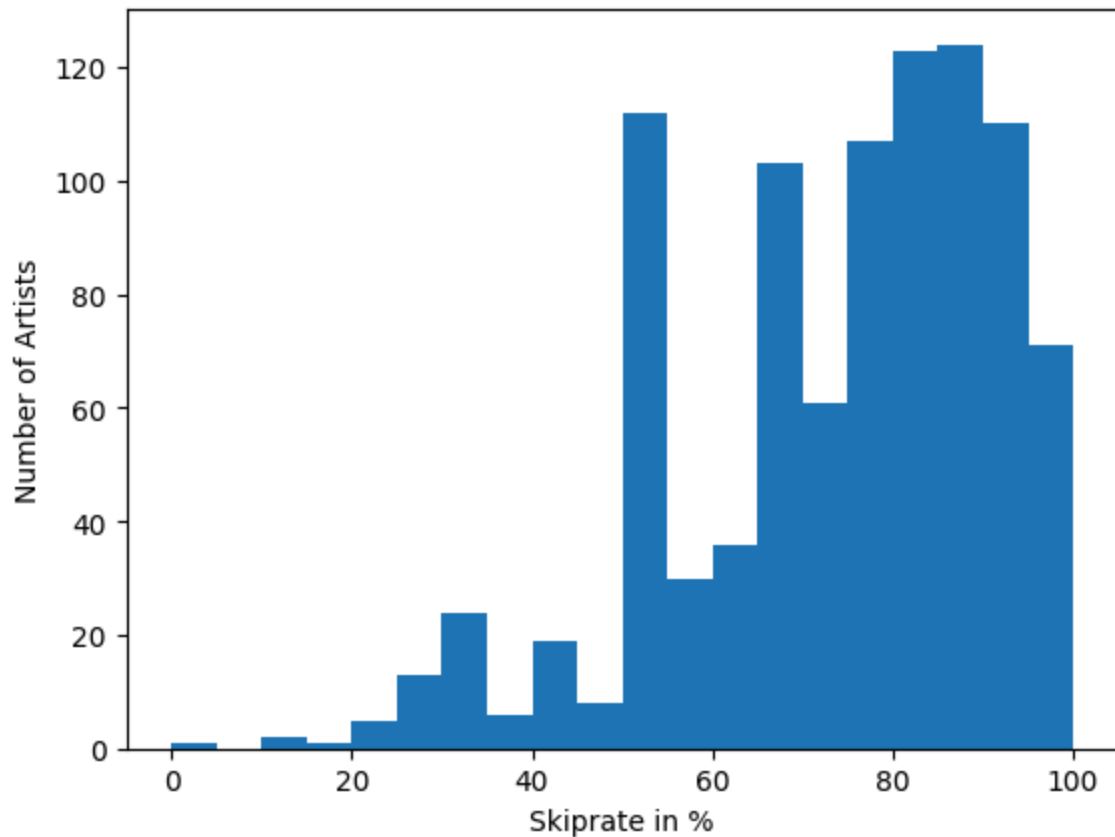
Create a histogram of the distribution of the skip-rate values of the different artists in your DataFrame `df_skipped`, with skip rates on one axis and number of artists on the other.

Then, retrieve the skip rate for Ariana Grande from your DataFrame `df_skipped`. Run the code in the cell below. Where on this distribution does Ariana Grande fall? Do I skip her songs more than average, or less?

```
In [25]: df_skipped["skipRate"].plot(kind="hist", bins=20)

plt.xlabel("Skiprate in %")
plt.ylabel("Number of Artists")
```

```
Out[25]: Text(0, 0.5, 'Number of Artists')
```



```
In [26]: print("skiprate:", df_skipped.loc["Ariana Grande", "skipRate"], "\n")

print("Mean skiprate", df_skipped["skipRate"].mean(), "\n")

# Looking at the mean skiprate of all the artists combined, Ariana is skipped more

print(top_10_artists)
```

skiprate: 99.52939959662822

Mean skiprate 73.04822293282288

```
artistName
Ariana Grande      178996.003
Joi                137229.562
The Pretty Reckless 110293.430
Cage The Elephant   95587.575
Lana Del Rey        90543.113
Greta Van Fleet     88026.405
The Neighbourhood    85673.375
Gorillaz            84858.371
Led Zeppelin        77030.802
Arctic Monkeys      76444.236
Name: secPlayed, dtype: float64
```

Part 4: Questions

Q1: Did I skip a lot of Ariana Grande's songs, or did I not, compared to the rest of the dataset?

According to dataframe analysis you skipped roughly 99.5% (practically all) of Ariana Grande songs.

Q2: What might be some possible reasons for Ariana Grande to be my nr.1 artist?

According to our conditions we mark a song as skipped when it is listened to for less than 30 seconds. When we look at the top 10 artist by playtime we can see that Ariana Grande takes first place, with a total play time of 49 hours and 26 minutes (178006 seconds).

The owner of the spotify account is under high suspicion of having a guilty pleasure song / songs from Ariana Grande's library. They might constantly listen to only certain parts of her songs, or maybe they sang the chorus on repeat during a 49 hour long road trip.